

Clinical Study Summary Report (CSSR)

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1. Administrative & Study Identification

Full Study Title: A Phase 1, Open-label, Multicenter Study to Assess Safety, Tolerability, PK, and Efficacy of MK-1084 as Monotherapy and as Part of Various Combination Therapies in Participants With KRAS G12C Mutant Advanced Solid Tumors

Short Title/Acronym: A Study of MK-1084 in KRAS Mutant Advanced Solid Tumors (MK-1084-001) / KANDLELIT-001

Protocol Number: MK-1084-001

NCT Number: NCT05067283

Trial Status: Recruiting **Sponsor Name:** Merck

Investigational Product: MK-1084 (Calderasib) and combination therapies (including pembrolizumab, pemetrexed, and fluorouracil)

Phase of Development: Phase 1

Medical Monitor: Merck Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the safety and tolerability of MK-1084 (calderasib) as monotherapy and as part of various combination therapies in participants with KRAS G12C mutant advanced solid tumors. **Secondary Objectives:** To evaluate the objective response rate (ORR), duration of response (DOR), and pharmacokinetics (PK) evaluated as mean plasma concentration of MK-1084.

2.2 Background & Rationale To address the critical unmet medical need in patients with advanced solid tumors harboring a KRAS G12C mutation. KRAS is one of the most prevalent oncogene mutations in human cancers, promoting uncontrolled tumor growth. MK-1084 (calderasib) is a selective KRAS G12C covalent inhibitor that has demonstrated promising antitumor activity by attenuating the KRAS signaling pathway. This study evaluates its safety and preliminary efficacy alone and combined with other therapies (such as immunotherapy and chemotherapy) to improve clinical outcomes and overcome resistance mechanisms.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Multi-arm (Dose Escalation & Expansion, Arms 1-6) **Blinding:** Open-label **Randomization:** Non-Randomized

3.2 Planned Enrollment & Duration Target \$N\$: 830 participants
Number of Sites: Global Multicenter **Estimated Study Start:** 17-Dec-2021 **Estimated Primary Completion:** 25-Feb-2030

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years. 2. Has measurable disease by RECIST 1.1 criteria. 3. Has adequate organ function. 4. Histologically or blood-based confirmation of KRAS G12C mutation in locally advanced unresectable or metastatic solid-tumor malignancy (including NSCLC and CRC). NSCLC participants may require specific prior treatment lines or histologic confirmation of PD-L1 TPS $\geq 1\%$ depending on the study arm. CRC participants must have had previous treatment failure of 1 or 2 lines of systemic therapy. 5. Male participants agree to protocol-specified contraception requirements including refraining from donating sperm. Female participants must not be pregnant or breastfeeding, must agree to contraceptive requirements, and must have a negative highly sensitive pregnancy test within 24 hours (urine) or 72 hours (serum) before the first dose.

4.2 Exclusion Criteria 1. Clinically active central nervous system (CNS) metastases and/or carcinomatous meningitis. 2. Has received chemotherapy, definitive biological cancer therapy, or definitive radiation within 4 weeks (2 weeks for palliative radiation) before first dose of study intervention. Has received a live or live-attenuated vaccine within 4 weeks of study start. 3. Active infection requiring systemic therapy, known history of HIV infection, Hepatitis B (HBV) virus, or known active Hepatitis C (HCV) virus infection. 4. Has a history of a second malignancy, unless potentially curative treatment has been completed with no evidence of malignancy for 5 years. 5. Has a history of interstitial lung disease, noninfectious pneumonitis requiring active steroid therapy, ongoing pneumonitis, or an active autoimmune disease requiring systemic therapy. 6. Has not fully recovered from any effects of major surgical procedure without significant detectable infection. 7. Ophthalmological findings: intraocular pressure >21 mm Hg, glaucoma, central serous retinopathy, retinal vein/artery occlusion, and/or retinal degenerative disease. 8. **Arm 4 Only:** Unable to interrupt NSAIDs/ aspirin (other than ≤ 1.3 g/day) before, during, and after pemetrexed administration; or is unable/unwilling to take folic acid, vitamin B12, and dexamethasone.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Participants receive MK-1084 daily oral escalating doses. Depending on the assigned arm, MK-1084 is given as monotherapy or in combination with other agents such as pembrolizumab, pemetrexed (requiring vitamin supplementation), or fluorouracil-based regimens (e.g., for CRC). **Route of Administration:** Oral (MK-1084) and Intravenous (Combination therapies). **Duration of Treatment:** Until progressive

disease or discontinuation (up to approximately 35 cycles or 24 months).

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for advanced solid tumors. **Prohibited:** Concurrent systemic anticancer therapies, live or live-attenuated vaccines within 4 weeks of study start, and active immunosuppressive therapy.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, KRAS G12C/PD-L1 Verification, Baseline Tumor Imaging, Highly Sensitive Pregnancy Test
Baseline	Day 0	Enrollment, First Dose (MK-1084 +/- Combo Therapy)
Interim	Every 3 Weeks	Safety Labs, Efficacy Assessment, Tumor Imaging (CT/MRI every 6 weeks for the first 18 weeks)
End of Study	30 Days Post-Tx	Final Vital Signs, AE Monitoring (serious AEs tracked up to 90 days)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints

- 1. Number of Participants Who Experience a Dose-Limiting Toxicity (DLT):** Defined by CTCAE 5.0, evaluated for type, severity, onset, and resolution.
- 2. Number of Participants Who Experience an Adverse Event (AE):** Any unfavorable/unintended sign, symptom, or disease temporally associated with the use of a medicinal product.
- 3. Number of Participants Who Discontinue Study Treatment Due to an AE.**

7.2 Secondary Endpoints

- 1. Objective Response Rate (ORR):** Percentage of participants achieving Complete Response (CR) or Partial Response (PR) per RECIST 1.1.
- 2. Duration of Response (DOR):** Time from first documented evidence of CR or PR until progressive disease per RECIST 1.1.
- 3. Pharmacokinetics:** Mean Plasma Concentration of MK-1084.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: AEs evaluated from baseline through 30 days after cessation of study treatment. **Serious Adverse Events (SAEs):** Evaluated and reported continuously (up to 90 days post-treatment). **Data Monitoring Committee (DMC):** Internal safety monitoring and dose-escalation review managed by Merck.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: All participants who receive ≥ 1 dose of study treatment are included in the safety and efficacy analyses. **Sample Size Justification:** Study enrollment target of 830 patients powered for a standard Phase 1 dose-escalation and multi-arm cohort expansion design. **Handling of Missing Data:** Standard oncology approaches for missing data; participants without documented progression are censored at their last evaluable tumor assessment.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Standard phase 1 onsite and remote monitoring of trial sites evaluating dose-limiting toxicities. **Audit Trail:** eCRF locking, tracking of protocol deviations, and source data verification.

11. Study Identifiers & Governance

Primary ID: MK-1084-001 **Registry Number:** NCT05067283 **Sponsor/Lead Organization:** Merck **Study Title:** A Study of MK-1084 in KRAS Mutant Advanced Solid Tumors (MK-1084-001) **Official Regulatory Title:** A Phase 1, Open-label, Multicenter Study to Assess Safety, Tolerability, PK, and Efficacy of MK-1084 as Monotherapy and as Part of Various Combination Therapies in Participants With KRAS G12C Mutant Advanced Solid Tumors

12. Clinical Framework (Oncology Specific)

Primary Condition: KRAS G12C Mutant Advanced Solid Tumors (including NSCLC and CRC) **Diagnosis Threshold:** Confirmed KRAS G12C mutation via validated assay (histologically or blood-based) **Medical Methodology:** Targeted covalent KRAS G12C inhibition +/- Combination Therapies **Optimization Target:** Tumor volume reduction, safety validation, and objective response achievement

13. Study Procedures & Methodology

Management Pathway: Standard clinical oncology pathway for advanced solid tumors **Education Protocol (HCP):** Investigator meetings and Phase 1 safety/DLT reporting training **Education Protocol (Patient):** Informed consent process, dosing instructions, and strict contraception/pregnancy testing mandates **Quality Audit Mechanism:** Clinical site monitoring and internal sponsor safety review committee

14. Observation & Follow-Up Schedule

Total Duration: Up to ~24 months (35 cycles) of active treatment, followed by survival follow-up **Onsite Visit Cadence:** Every 3 weeks (Q3W) synchronized with combination therapy administration **Remote Monitoring Frequency:** Inter-cycle telehealth safety check-ins as applicable **Checklist Review Interval:** Tumor imaging (CT/MRI) assessed every 6 weeks up to week 18, every 9 weeks for the remainder of year 1, and every 12 weeks thereafter

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---	
Principal Investigator	[Name]	_____	03-Apr-2026					
Clinical Operations Lead	[Name]	_____	03-Apr-2026		Lead			
Statistician	[Name]	_____	03-Apr-2026					